OXIDATION REDUCTION REACTIONS

Oxidation Reduction reaction is also known as —REDOX‖ reaction. It comprises of both

oxidation and reduction reactions. Read various definitions of oxidation and reduction reactionsr

Definitions of Oxidation and Reduction

a. Oxidation and Reduction in terms of Oxygen.

Oxidation is the addition of oxygen and reduction is the removal of oxygen.

i. Burning of magnesium in oxygen

$2Mg_{(s)} + O_{2(g)}$

$2MgO_{(s)}$

Conversion of magnesium metal to its oxide by the addition of oxygen is Oxidation. Magnesium

is oxidized.

ii. Passing hydrogen gas over heated copper(ii) oxide: $H_{2(g)}$ + CuO to to Cu, by the removal

of oxygen is reduction. CuO is reduced to copper metal by hydrogen gas.

$H_{2(g)}CuO$

$Cu + H_2O$

(b) Oxidation and Reduction in terms of hydrogen: Oxidation is the removal of hydrogen

from a substance, while reduction is the addition of hydrogen to a substance e.g.

i. Decomposition of ammonia:

$2NH_{3(g)}$

$N_{2(g)} + 3H_{2(g)}$. The conversion of NH to N by the removal of hydrogen

3 2

is oxidation. Ammonia is oxidized to nitrogen.

ii. Formation of ammonia:

$N_{2(g)} + 3H_{2(g)}$

$2NH_{3(g)}$. The conversion of N_2 to NH_3 by the addition of H_2 is

reduction. Nitrogen is reduced to ammonia by the addition of H_2 gas.

c. Oxidation and Reduction in terms of Electronegativity of elements.

Oxidation is the addition of an electronegative element to another element, while reduction is

the removal of an electronegative element from a compound. Oxidation and reduction involve all

the elements, either as being electro-negative or electro-positive. In the periodic table,

electropositive elements are on the right, while those on the left are electronegative.

hydrogen. HCl is reduced to H_2 gas.

b. Oxidation and Reduction in terms of Electron transfer

Oxidation is the process of electron loss; Reduction is the process of electron-gain. This concept

of oxidation and reduction is limited to reactions between metals and non-metals, where

electrons are transferred from the valence shell of the metal atom to the valence shell of the non-

metal atom, leading to the formation of electrovalent or ionic compounds. Magnesium atom

looses its two valence electrons to form magnesium ion.

$Mg - 2e^-$

Mg^{2+}

Mg is oxidized to Mg^{2+}.

ii. When sulphur (non-metal) gains two electrons in order to complete its valence shell, it forms

sulphur ion.

$S + 2e^-$

S^{2-}

Sulpur atom is reduced to its ion.

e. Oxidation and Reduction in terms of Oxidation Number (O.N.)

Oxidation is an increase in oxidation number. Reduction is a decrease in oxidation number. The

oxidation number of an element is the charge on its atom or ion. The oxidation number of a

neutral atom is zero, because the number of protons in its nucleus is equal to the number of

electrons in its shell.

294

Mg

Mg^{2+}

O

+2

The oxidation number of Mg increases from 0 to +2. When sulphur atom ionizes by gain of two

electrons; it forms sulphur ion.

S

S^{2-}

O

-2

The oxidation number of sulphur ion, S^{2-} is -2. Hence the O.N. of sulphur decreases from 0 to -2.

It is reduction.

Oxidation No of Central Elements in Some Compounds

Rules for Assigning Oxidation Numbers (O.N.)

1.

The O.N. of an atom in an uncombined free element, whether monatomic or polyatomic

is zero; e.g. Noble gases (He, Ne etc.), metals (Na, Cu, etc.), solid non-metals (C, S, etc.),

and molecules such as (H_2, O_2, Cl_2, P_2, S_2 etc.).

The O.N. of one oxygen atom is -2 in its compounds, (except in peroxides: H_2O_2, Na_2O_2

or O^{2-}, where it is -1 and in its binary compound with fluorin, OF_2 where it is +2.

The oxidation number of one hydrogen atom is +1 in its compounds (except in hydrides

of metals, e.g. NaH, where it is -1).

Group I alkali metals (Li, Na, K, etc.) have O.N. of +1, group II alkaline - earth metals

(Mg, Ca, etc.) have O.N. of +2, while group III metals (Al, etc.) have O.N. of +3 in all

their compounds.

The halogens (F, Cl, Br, I, etc.) in group VII have O.N. of -1 per atom in their binary

compounds only (except when bonded to a more electronegative atom - as in IF_7, where

the O.N. of iodine atom is +7.

The O.N. of the atom in a monatomic ion is equal to the charge on it

sodium atom in Na is +1; that of Al in Al is +3; and that of S in S is -2.

In any complex ion, the sum of the O.N's of all the atoms is equal to the charge on the

ion.

In any molecule or compound, the sum of the oxidation numbers of all the atoms adds to

zero.

Worked examples

Example 1:

Calculate the oxidation number of sulphur in tetraoxosulphate (vi) ion.

The oxidation number of SO_4^{2-} = -2

:. (oxidation number of S) + 4(oxidation number of O) = -2

oxidation number of S = 6

S + [4x(-2)] = -2

S-8 = -2

S = -2+8 = +6

Example 2:

Calculate the oxidation number of sulphur in K SO . Write the IUPAC nomenclature of the

2

4

compound.

K_2SO_4

(+1x2)+S+(-2x4) = 0

2 + S + (-8) = 0

2 + S -8

= 0

S2

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electropositive elements are on the right, while those on the left are electronegative.

In non-metals: P, C, S, I, Br, N, Cl, O, F

Increasing electronegativity

In metals and hydrogen:

Increasing electropositivity

K, Na, Ca, Mg, Al, Zn, Fe, Sn, Pb, H, Cu, Hg, Ag, Agu

Increasing electronegativity

i. Formation of Sodium Chloride:

$2Na_{(s)} + Cl_{2(g)}$

$2NaCl_{(s)}$. The conversion of sodium to sodium chloride by the

addition of chlorine is oxidation, because chlorine is more electronegative than sodium. Sodium

is oxidized to sodium chloride.

ii. Action of a reactive metal on and acid:

$Fe_{(s)} + 2HCl_{(aq)}$

$FeCl_{2(aq)} + H_{2(g)}$. The conversion of HCl to H_2 gas by the

removal of chlorine is reduction, because in HCl, chlorine is more electronegative than

hydrogen. HCl is reduced to H_2 gas.

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